CHEMISTRY EDUCATORS SOCIETY

SEMINAR QUESTIONS 2026

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FOREWORD

The **Chemistry Educators Society** is your companion to achieve excellent results. This seminar has been organized for you to interact with the most celebrated and knowledgeable facilitators of Chemistry. Their role is to guide you in the question approach and also equally to caution you about the mistakes that you make in the final examination which cost you excellent grades.

Nothing is more difficult and more precious than making the right decision, Kindly students, and always spare time to select questions where you can get maximum marks. Always do a comparison of the questions before responding.

Mulabbi Nicholas Elijah
(President CES)

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SEMINAR ITEMS

ASSESEMENT OBJECTIVE 3 (AO3)

The topics in this assessment objective include;

1. Thermochemistry
2. Reaction Kinetics
3. Electrochemistry

Item one:

During an environmental audit of an abandoned industrial metal refinery in Western Uganda, an environmental toxicology team discovered that runoff water was leaching into a downstream agricultural wetland. Local farmers reported severe illness among livestock and household members who drank from the marshland. Water samples collected from the site revealed high concentrations of two inorganic copper compounds: anhydrous copper(II) chloride and copper(II) sulfide. To pinpoint the source of poisoning, the toxicity board must identify which compound dissolves significantly in natural water bodies, as insoluble salts pose minimal immediate systemic danger. Furthermore, to treat poisoned patients, clinical pharmacologists are evaluating the intravenous administration of Ethylenediaminetetraacetic acid (EDTA). Effective dosage protocol design requires establishing the elimination kinetics, half-life, and rate constant of EDTA in human plasma.The investigating laboratory provided the following experimental datasets:

Data Set I

$Cu(s) \longrightarrow Cu(g)$		$+338\ kJ\ mol^{-1}$
$Cu(g) \longrightarrow Cu^{+}(g)$		$+745\ kJ\ mol^{-1}$
$Cu^{+}(g) \longrightarrow Cu^{2+}(g)$		$+1958\ kJ\ mol^{-1}$
$\frac{1}{2}Cl_2(g) \longrightarrow Cl(g)$		$+121\ kJ\ mol^{-1}$
$Cl(g) \longrightarrow Cl^{-}(g)$		$-349\ kJ\ mol^{-1}$
$Cu^{2+}(g) + S^{2-}(g) \longrightarrow$	$CuS(s)$	$-3506\ kJ\ mol^{-1}$
$Cu(s) + Cl_2(g) \longrightarrow$	$CuCl_2(s)$	$-220\ kJ\ mol^{-1}$
$Cu^{2+}(g) + (aq) \longrightarrow$	$Cu^{2+}(aq)$	$-2100\ kJ\ mol^{-1}$
$Cl^{-}(g) + (aq) \longrightarrow$	$Cl^{-}(aq)$	$-360\ kJ\ mol^{-1}$
$S^{2-}(g) + (aq) \longrightarrow$	$S^{2-}(aq)$	$-1050\ kJ\ mol^{-1}$

Data Set II

Concentration of EDTA ($moldm^{-3}$)	1.48	1.45	1.43	1.41	1.38	1.36
Time (hours)	1.0	2.0	3.0	4.0	5.0	6.0

Your chemistry teacher came across this information in one of the community meetings.

Task: As a student of Chemistry analyses and interpret the data for the community health team.

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Item two

A student innovation club at a secondary school in Uganda aims to construct a portable electrochemical cell capable of producing an electromotive force (e.m.f.) greater than **1.20 V**. The students plan to utilize a metal, **M**, paired with zinc (**Zn**). However, they are unsure if this combination will meet their voltage target. Upon consulting the school laboratory technician, they were provided with the following thermochemical and electrochemical data, along with a chemical equation representing the proposed displacement reaction:

$$ZnO_{(s)} + M_{(s)} \longrightarrow MO_{(s)} + Zn_{(s)}$$

Enthalpy of atomization of zinc, $\Delta H^{\theta}_{at}(\mathrm{Zn}) = +126\ \mathrm{kJ\ mol^{-1}}$

First ionization energy of zinc, $IE_1(\mathrm{Zn}) = +906\ \mathrm{kJ\ mol^{-1}}$

Second ionization energy of zinc, $IE_2(\mathrm{Zn}) = +1733\ \mathrm{kJ\ mol^{-1}}$

Bond dissociation energy of oxygen, $E(O = O) = +498\ \mathrm{kJ\ mol^{-1}}$

First electron affinity of oxygen, $EA_1(\mathrm{O}) = -142\ \mathrm{kJ\ mol^{-1}}$

Second electron affinity of oxygen, $EA_2(\mathrm{O}) = +731\ \mathrm{kJ\ mol^{-1}}$

Standard enthalpy of formation of metal oxide, $\Delta H^{\theta}_{f}(\mathrm{MO}) = -602\ \mathrm{kJ\ mol^{-1}}$

Lattice energy of zinc oxide, $U_L(\mathrm{ZnO}) = +4010\ \mathrm{kJ\ mol^{-1}}$

Faraday's constant, $F = 96,500\ \mathrm{C\ mol^{-1}}$

The technician noted that to address their concern, the students must determine the lattice energy enthalpy of formation of ZnO, evaluate the thermodynamic stability of Zinc oxide and feasibility of the reaction.

Item Three

A company focused on renewable energy is working on creating hydrogen gas to power fuel-cell charging stations, offering a clean way to replace fossil fuels. Before starting a small-scale plant, they're taking a close look at the chemical steps involved.

$$\mathbf{2\ X\ (aq) + Y\ (aq) \longrightarrow Products + H_2\ (g)}$$

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To be considered efficient, the reaction must have a rate constant exceeding 2.0×10^{-2} $mol^{-1}dm^3$ and a standard enthalpy of reaction which is exothermic in a magnitude greater than -150 $kjmol^{-1}$. A series of experiments were carried out by the technicians employed by the company and the results are indicated in the table below

Experiment	[X] $moldm^{-3}$	[Y] $moldm^{-3}$	Initial rate ($moldm^{-3}s^{-1}$)
1	0.10	0.10	2.0×10^{-3}
2	0.20	0.10	8.0×10^{-3}
3	0.30	0.10	18.0×10^{-3}
4	0.10	0.20	4.0×10^{-3}

Looking at chemistry data books, the technicians also were able to access the enthalpy of formation of X, Y and the products in the reaction of interest. The thermochemical data was

Species	Enthalpy of formation ($kjmol^{-1}$)
X	-120
Y	-250
Products	-670

equally presented in a table

The technicians are eager to investigate the factors that influence the rate of hydrogen production. The company intends to establish production facilities in remote locations, prioritizing the seamless operation, safety, and environmental sustainability of the facilities.

Task

a) Analyze the data provided and utilize it to ascertain whether the reaction can be employed for electricity generation and assist the company in evaluating the efficiency of the production process.

b) Please explain the factors that may influence the production rate of the fuel.

c) Evaluate the significance of the principles used in real-world experiments and provide recommendations on the quality of fuel that the company should consider.

Item Four

The technical maintenance team at the Lake Victoria Water Infrastructure Project in Jinja district noticed that underground iron distribution conduits undergo accelerated corrosion when conveying acidic runoff. The overall corrosion process is represented by the equation:

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$$2Fe(s) + O_2(g) + 4H^+(aq) \longrightarrow 2Fe^{2+}(aq) + 2H_2O(l)$$

To determine the exact influence of acidity on corrosion speed and evaluate the feasibility of the reaction, a research group of Advanced Level Chemistry students conducted kinetic investigations at 25°C. They gathered the following rate data alongside relevant standard electrode potentials:

Experiment	$[H^+]$ (mol L^{-1})	Initial Rate (mol $dm^{-3}s^{-1}$)
1	0.10	2.5×10^{-6}
2	0.20	5.0×10^{-6}
3	0.40	1.0×10^{-5}

$Fe^{2+}(aq)/Fe(s) : E^\circ = -0.44\ V$

$O_2(g)/H_2O(l)/H^+(aq) : E^\circ = +1.23\ V$

Student Inquiries & Technical Requests:

1. **Acidity Analysis:** One student wants to verify whether the water sample experienced an increase in hydrogen ion concentration when a new trial produced an initial reaction rate of $2.0x10^{-5}$ mol $dm^{-3}s^{-1}$.

2. **Corrosion Prevention Strategy:** The chief engineer provided an additional electrode potential value, and asked the team to explain how this data can be applied to protect the pipeline network.

$$Zn^{2+}(aq)/Zn(s) : E^\circ = -0.76\ V$$

3. **Kinetic Factors:** The engineer also requested an explanation of two other reaction conditions that influence the rate of corrosion in metallic pipes.

Task

As the student team leader, you are are supposed to prepare a clear, structured technical report of your team's findings and solutions to present to the project engineer.

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Item five

The effectiveness of Calcium iodide used in making hand warmers depends on the heat released when it dissolves in water, while its formation is determined by the electromotive force of the corresponding cell reaction.

At Namanve Industrial Park, a product-development team is testing two self hand warmers, one containing Calcium iodide and the other potassium iodide. Before launch, the team needs to understand the factors on which the lattice energy of the salt being tested depends, the effectiveness of the best salt in making handwarmers and whether Calcium iodide can be made from calcium and iodine by a redox reaction in an electrochemical cell. They researched and found the following data;

The thermochemical data;

Enthalpy change	**($kJmol^{-1}$)**
Enthalpy of atomization of iodine	106.8
First ionization energy of calcium	590
Second ionization energy of calcium	1145
Enthalpy of atomization of calcium	178.2
Enthalpy of formation of calcium iodide	-533.5
Electron affinity of iodine	-295.4
Lattice energy of KI	-636
Hydration energy of K^+, Ca^{2+}, I^- respectively	322, 1650, 295

The electrochemical data;

Ca^{2+} / Ca	-2.87
I_2 / I^-	0.54

The team was unable to use this data as their technical person was not around and they were referred to you for help in order to make informed decisions.

Task; As a learner of chemistry;

Help the team to analyze the data and make recommendation and address their other concerns.

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ASSESEMENT OBJECTIVE 4 (AO4)

The topics in this assessment objective include;

1. Equilibria I
2. Equilibria II
3. Moles and equations

Item one

A local quality control laboratory in Mukono recently received an unlabelled shipment of a solid organic compound suspected to be **naphthalene**. To confirm its identity, the quality assurance team needs to accurately determine its relative molecular mass using colligative property measurements. A group of Senior 6 Chemistry students on an industrial attachment program was tasked with analyzing the sample. They dissolved 0.142 g of the pure solid naphthalene in 20.25 g of pure liquid benzene in a sealed apparatus. Using a high-precision calibrated digital thermometer, they measured the freezing points of both the pure solvent and the resulting solution:

Freezing point of pure benzene: $5.50^\circ C$

Freezing point of naphthalene-benzene solution: $5.216^\circ C$

(Cryoscopic constant, K_f, of benzene = $5.12^\circ C\ kg\ mol^{-1}$)

A Senior 5 student observing the experiment was curious about the reason for freezing point depression, how the numerical data can be manipulated to obtain the molecular mass of the solute, and the limitations of using this method for molecular mass determination. While reviewing literature on benzene, the student also noted that because benzene is volatile and toxic, chemical industries often hydrogenate it into cyclohexane over a palladium catalyst before disposal or further processing:

$$C_6H_6(l) + 3H_2(g) \rightleftharpoons C_6H_{12}(l) \quad \Delta H = -208\ kJ\ mol^{-1}$$

In a bench-scale simulation performed at 298 K, 7.8 g of benzene and 0.6 g of hydrogen gas were sealed in a 2.0 litre reaction vessel with a palladium catalyst. At equilibrium, 0.08 moles of unreacted benzene remained. The Senior 5 student wants to know how to calculate the equilibrium constant Kc from this data and what operational conditions would maximize the conversion (elimination) of toxic benzene.

Task

Write a report to address the issues raised in the scenario

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Item two

A comprehensive understanding of chemical equilibrium is essential for optimizing conditions in industrial processes while mitigating environmental impact. A fertilizer manufacturing company is currently grappling with the threat of closure due to public outcry regarding the release of toxic fumes into the atmosphere.

An environmental analysis revealed that burning a 50.05g of sulphur yields 100.00g of the toxic fumes, Q, released from the plant.The molar mass of Q is 64. To mitigate the effect of the emissions, the Quality Assurance Manager suggested converting gas Q into a less hazardous substance, P, based on the equilibrium reaction:

$$2\,Q\,(g) + O_2\,(g) \rightleftharpoons 2P\,(g)$$

During initial testing:

3.0 moles of gas Q and 1.5 moles of oxygen were introduced into a closed reaction vessel and heated to 400°C. At equilibrium, the mixture contained 0.5 moles of remaining oxygen gas.When the temperature was raised to 500°C, the equilibrium mixture was found to contain 40% of the initial moles of gas Q.

The company management demands a comprehensive technical assessment of the chemical identity of Q, the toxic fumes, and the factors that can facilitate the conversion of these fumes into the less toxic substance, P. Additionally, the company is interested in understanding the application of the aforementioned reactions that they have attempted to investigate.

Task

As a chemistry student, analyze the provided experimental data and prepare a comprehensive report addressing each of the management's concerns.

Item three

A beverage manufacturing plant extracts subterranean water from limestone-rich aquifers for bottled mineral water production. The source water naturally contains dissolved calcium hydrogencarbonate, formed when atmospheric carbon dioxide dissolves in groundwater and reacts with limestone strata. Following reports from consumers regarding a white cloudy suspension forming in stored bottles, especially those exposed to warmer temperatures, the quality assurance department suspected that temperature-induced shifts in ionic equilibria were causing calcium carbonate precipitation. To evaluate the extent of the problem under varying storage conditions, laboratory tests were conducted to determine the concentrations of key dissolved ions in three distinct bottled water batches:

Batch Sample	$[Ca^{2+}]$ (mol dm^{-3})	$[CO_3^{2-}]$ (mol dm^{-3})
A	1.0×10^{-2}	2.0×10^{-7}
B	2.5×10^{-2}	3.0×10^{-7}
C	4.0×10^{-2}	5.0×10^{-7}

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Solubility product (K_{sp}) of calcium carbonate at test temperature = 4.0×10^{-9} $mol^2 dm^{-6}$

The company requires expert chemical advice to establish why the precipitate forms and to recommend optimal storage conditions that prevent quality deterioration. In a separate quality-improvement trial prior to bottling, the plant is testing a solvent extraction procedure to remove trace amounts of an organic impurity, **X**, from the raw water supply.

The partition coefficient KD for contaminant **X** between an organic extraction solvent and water at 25°C is 4.0. A 100cm³ batch of raw water containing 0.80g of **X** is extracted twice in succession using 25 cm³ of the organic solvent for each step, allowing chemical equilibrium to establish fully during each stage

TASK

a) Perform ionic product calculations for Samples A, B, and C to identify and explain which batch(es) will develop visible precipitates during storage.

b) Calculate the mass of organic contaminant **X** remaining in the aqueous layer following the two-step extraction process, and assess the efficiency of this method for industrial water purification.

c) Detail the factors influencing carbonate precipitation in the bottled water, and discuss how principles of equilibria are applied across industrial and real-world processes.

Item four

Steam distillation is a key separation technique widely utilized in the pharmaceutical and natural product industries to extract temperature-sensitive, immiscible organic compounds.An essential oils company is utilizing steam distillation to isolate an active medicinal agent, **Compound Z**, from a botanical extract. The resulting distillate consists of a two-phase mixture of water and organic Compound Z. Laboratory analysis of the distillate yields the following vapor pressure data at varying temperatures:

Temperature(°C)	20.0	40.0	60.0	80.0	100.0
Vapour pressure of water (atm)	0.22	0.26	0.30	0.35	0.39
Vapour pressure of Z (atm)	0.35	0.42	0.49	0.56	0.63

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At a collection temperature of **97 °C**, the resulting distillate yields **4.2 g of Compound Z** and **1.1 g of water**.

In addition to extraction, the company operates an industrial ammonia plant using the Haber process. During a test trial, a **4.0 dm3** reaction vessel is charged with **2.5 moles of nitrogen gas** and **7.5 moles of hydrogen gas**. At equilibrium, **1.6 moles of ammonia** are produced:

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} \quad \Delta H = -92 \text{ kJ mol}^{-1}$$

Target Criteria

The company's technical advisory guidelines specify that for optimal industrial efficiency:

1. The extracted organic compound should have a molecular mass **greater than 32 g mol^{-1}**.
2. The equilibrium constant (Kc) for the ammonia synthesis reaction should be **greater than 0.15 mol^{-2}dm^{6}**

Tasks

As a chemistry specialist, assist the chemical engineers by addressing the following tasks:

i. **Extraction Efficiency Analysis**
ii. **Ammonia Yield Optimization**
iii. **Real-Life Applications:** Discuss other industrial and practical applications of steam distillation and dynamic chemical equilibrium

Item five

Fertilizers containing inorganic salts such as nitrates, chlorides, sulfates, and phosphates are commonly applied to farmland to enhance crop yields. However, excessive or prolonged use can alter soil chemistry and adversely affect plant growth. GreenFields Agro-Enterprise specializes in cultivating sorghum and cabbages, both of which require alkaline soil conditions to achieve optimal yield. For several seasons, the farm manager consistently applied ammonium nitrate fertilizer to boost soil productivity. Recently, crop yields have dropped significantly despite increasing the quantity of fertilizer applied. Concerned by this trend, the manager suspects that the soil acidity has increased due to the fertilizer. To test this hypothesis, a soil analysis was conducted:

K_b of ammonia at $25°C = 1.85 \times 10^{-5}$ mol dm^{-3}

A 26.4 g soil sample treated with ammonium nitrate was dissolved in 500 cm^3 of distilled water and filtered.

25.0 cm^3 of the resulting filtrate required 10.0 cm^3 of 0.05 M sodium hydroxide solution for complete neutralization using methyl orange indicator.

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Additionally, an agricultural extension officer advised the manager that spraying sorghum fields with an aqueous mixture of ammonia and ammonium chloride can maintain the soil pH at approximately 8.95, buffering it against rapid pH changes. The farm has run out of the commercial mixture but has access to low-cost stock solutions of ammonia and solid ammonium chloride. The manager requires technical guidance on how to prepare this buffer locally.

Task: As a Chemistry learner, prepare a report to assist the farm manager in which you:

(a) Use the provided analytical data to calculate the soil pH, determine the concentration of hydrogen ions, and justify why crop yields declined under the current fertilizer regimen.

(b) Explain the chemical principles behind preparing the buffer solution locally using ammonia and ammonium chloride to resolve the shortage.

(c) Outline other practical, real-life applications of chemical systems that function similarly to the ammonia–ammonium chloride buffer mixture.

Item six

A chemical manufacturing facility in Uganda processes a liquid mixture of methanol and cyclohexane, attempting to separate and recover both solvents using simple distillation. However, the process yields low-purity solvents, resulting in significant material loss. Quality control measurements recorded the boiling points and phase compositions for the mixture as follows:

Pure Cyclohexane: Boiling Point = 81°C

Pure Methanol: Boiling Point = 65°C

Mixture compositions and boiling points:

- At 70°C: Mole fraction of methanol is 0.27 in liquid, 0.12 in vapour.
- At 60°C: Mole fraction of methanol is 0.47 in liquid, 0.13 in vapour.
- At 55°C: Mole fraction of methanol is 0.56 in liquid, 0.50 in vapour.
- At 57°C: Mole fraction of methanol is 0.69 in liquid, 0.82 in vapour.
- At 61°C: Mole fraction of methanol is 0.83 in liquid, 0.94 in vapour.

Separately, the company produces a cleaning solution containing methanoic acid (Ka = 3.75). During quality testing, 50.0 cm^3 of 0.050 mol dm^{-3} methanoic acid was mixed with 50.0 cm^3 of 0.020 mol dm-3 sodium hydroxide at 25°C. During storage, 5.0 cm^3 of 0.020mol dm^{-3} hydrochloric acid was accidentally introduced into the mixture. The company needs to determine if this solution functions effectively as a buffer and what practical uses similar systems have.

Tasks

(a) Analyze the laboratory distillation data to explain the root cause of the poor separation and solvent losses.

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(b) Evaluate whether the methanoic acid mixture successfully maintains a stable pH despite the addition of acid, and explain other practical applications of similar buffer systems.

Item Seven

A bottled mineral water company sources its water from underground limestone layers. The water naturally contains dissolved calcium hydrogen carbonate formed through the interaction of rainwater, carbon dioxide, and limestone rocks.

Recently, customers have complained that some bottled water develops a white precipitate after prolonged storage, particularly in bottles stored in warm environments. The company's quality assurance team suspects that changes in the equilibria involving calcium carbonate are responsible.

The company carried out investigations under different storage conditions.

Table 1: Ionic Concentrations in Three Bottled Water Samples

Sample	$[Ca^{2+}]$ (mol dm^{-3})	$[CO_3^{2-}]$ (mol dm^{-3})
A	1.0×10^{-3}	2.0×10^{-3}
B	2.5×10^{-3}	3.0×10^{-3}
C	4.0×10^{-3}	5.0×10^{-3}

For calcium carbonate, $K_{sp} = 4.0 \times 10^{-9}$

The company wants to determine the causes of precipitate formation and identify suitable storage conditions that maintain water quality but lack expertise.

To improve water quality, the company is also investigating the use of a solvent extraction process to remove traces of an organic contaminant, **X**, from the water before bottling.

The distribution coefficient of contaminant **X** between an organic solvent and water at 25°C is 4.0. A 100 cm^3 sample of contaminated water containing 0.80 g of **X** is shaken with two successive portions of 25 cm^3 of the organic solvent and allowed to reach equilibrium.

Task

a) Analyse the data given for Samples **A, B** and **C** and use it to identify and explain the causes of precipitate formation and identify suitable storage conditions that maintain water quality and develop visible deposits during storage.

b) Determine the amount of contaminant **X** remaining in the aqueous phase after extraction and interpret the significance of your results in relation to water quality and purification.

Item seven

A food-processing company in Uganda manufactures fruit-based products that require controlled acidity for acceptable taste and shelf life. The company uses organic acids as acidity regulators and prepares buffer solutions to maintain the pH of its products during processing and storage.

During a school-industry study program, Senior Six learners joined the company's quality-control team when a consignment of an organic acid, HA, arrived with an incomplete label. Before using the acid, the company needed to confirm its identity, determine the amount of acid available and establish whether it could be used to prepare a suitable buffer.

The laboratory team established that HA is a saturated monocarboxylic acid with the general formula $C_nH_{2n}O_2$. To determine its molecular formula, 10.0 cm^3 of HA vapour was mixed

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with 70.0 cm^3 of oxygen and completely burnt. After cooling to room temperature, the gaseous mixture occupied 50.0 cm^3. When the mixture was passed through aqueous potassium hydroxide, its volume decreased to 30.0 cm^3. All gas volumes were measured under the same temperature and pressure conditions.

$$C_nH_{2n}O_2(g) + \frac{3n-2}{2} O_2(g) \rightarrow nCO_2(g) + nH_2O(l)$$

In another test, 25.0 cm^3 of a solution prepared from the consignment required 20.00 cm^3 of 0.1 M NaOH for complete neutralization. The company has 500 dm^3 of this acid solution available for production.

$$HA(aq) + NaOH(aq) \rightarrow NaA(aq) + H_2O(l)$$

The company intends to use part of the acid to prepare a buffer for a fruit product whose required pH is 4.60-4.80. A technician proposed mixing 0.1 M of HA with 0.080 M of NaA and making the mixture up to 1.00 dm^3.

For the acid identified from the combustion investigation: $Ka = 1.8 \times 10^{-5}$ mol dm^{-3}.

During processing, small quantities of acid from fruit ingredients may enter the product. The technical team explained that an effective buffer should resist large changes in pH when small amounts of acid or alkali are introduced. The company therefore needs to understand how the relative amounts of **HA** and $\mathbf{A^-}$ influence the buffer pH and why the buffer is more effective when appreciable quantities of both components are present.

As a member of the learner team, you have been asked to use the laboratory information and relevant chemical principles to advise the company.

Tasks

(a) Analyze the eudiometric and stoichiometric information to establish the identity and concentration of **HA**, and determine the amount of **HA** available in the production batch.

(b) Evaluate the proposed buffer formulation, explaining how its composition controls pH and assess its suitability and application in the company's fruit-processing operation.

Item eight

Industrial process utilize physical properties of some substances in preserving some of their products. In one of the manufacturing industries the chemical engineer is interested in formulating a mixture for cooling and preserving some of their products. They proposed to use an aqueous solution containing substance X. Before using the substance, they had to carry out some tests in order to confirm its purity basing on its melting point and molar mass which is approximately 101 g. Data analysis on substance X indicated that;

- Melting point is 334°C
- 3.04 g of substance X when dissolved in 280 g of water caused a depression in freezing point of 0.2°C. On cooling the mixture, the following results were obtained.

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Temperature (°C)	-14	-35	63	136.5	227.5	283.5
Percentage composition of the substance	7.5	18.75	40	56.26	76.25	88.75

In their view they wanted to use a mixture which had lowest temperature, and lower than -30.0°C. In order to obtain the desired mixture, they decided to cool the aqueous solution containing between 50 to 70% of substance X from a temperature of 245°C to -60°C.
Melting point and freezing point constant of water are 0°C and 1.86°C per mole per 1000g respectively.
You have been approached for some guidance.

TASK; As a learner of chemistry,

(a) Analyze the data obtained by the chemical engineer.

(b) Asses:

(i) The cooling process

(ii) The effects of the substance on physical properties of water mentioned in the scenario.

(c) Suggest applications of the lowest melting point mixture system.

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ASSESEMENT OBJECTIVE 1 (AO1)

The topics in this assessment objective include;

1. **Atomic, electronic structure and radioactivity**
2. **Bonding and structure**
3. **Periodicity I**
4. **Periodicity II**

Item one

During a field study at a chemical synthesis and manufacturing plant in Namanve Industrial Park, a group of Senior Five students investigated the periodic trends and chemical behavior of Period 3 elements and their compounds.The plant chief analyst presented data on the **first ionization energies** of Period 3 elements, highlighting distinct anomalies across the period. In the laboratory session, the students observed four white solid oxides of the first four Period 3 elements: **Sodium oxide**, **Magnesium oxide**, **Aluminum oxide**, and **Silicon dioxide**. The analyst demonstrated their reactions when separately treated with **dilute hydrochloric acid** and **sodium hydroxide solution**.The environmental and safety officer emphasized that while these elements and their oxides have extensive applications they could pose severe risks.The chemistry teacher asked the students to write a detailed technical report addressing the periodicity, chemical reactions, industrial applications, hazards, and safety mitigations associated with these Period 3 substances.

Item two

A team of Senior Five chemistry students visited an advanced metallurgy and aerospace materials research facility in the Namanve Industrial Zone. During the tour, the chief metallurgist demonstrated the unique physical and chemical properties of **Beryllium** and **Aluminum**.Although Beryllium belongs to Group II (Period 2) and Aluminum belongs to Group III (Period 3) of the periodic table, the researcher highlighted that they exhibit remarkably similar chemical behavior—a phenomenon known as a **diagonal relationship**. In the laboratory session, the students observed the reactions of metallic Beryllium and Aluminum samples when treated separately with **dilute hydrochloric acid** and **hot concentrated sodium hydroxide solution**.The safety and environmental officer noted that while both metals and their alloys/compounds are valuable but equally pose risks to the environment and health.The chemistry teacher instructed the students to draft a comprehensive scientific and safety report addressing the diagonal relationship between Beryllium and Aluminum, their chemical reactions, applications, associated hazards, and mitigation measures.

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Item three

The manufacturing industry's need for electricity is expanding quickly. A nuclear power plant has being developed in a big city to assist meet this need. In order to observe how the rays deflected, the technical team at the factory first channelled emissions from a radioactive source through an electric field. The group monitored the radioactive decay of lead-212 in a subsequent investigation. Lead (atomic number 82) was allowed to undergo a beta decay, followed by an alpha decay and finally a beta decay. In a third experiment, the team found that thorium-232 naturally decays, releasing a total of four beta and six alpha particles. In the fourth experiment, a sample of a radioactive isotope with a three-hour half-life was found to have an initial activity of 1000 counts per second.A comprehensive technical report including the following specifics is needed by the management team:the main physical characteristics of radioactive materials and the kinds of radiation they release. As lead decayed, the identities of the nuclides formed.The final stable isotope generated at the end of the decay sequence of thorium is identified. The sample's residual activity in the fourth experiment after two hours. The dangers to one's health and safety that come with using radiation in real-world situations.

Task

Write a thorough chemistry report that answers every query and issue brought up by the management team of the nuclear plant.

Item four

A group of Senior Six chemistry students visited a chemical processing plant in Mukono that manufactures disinfectants, water treatment chemicals, and household bleaches. During the facility tour, the industrial chemist demonstrated samples of the Group 17 elements (Halogens): **Fluorine**, **Chlorine**, **Bromine**, and **Iodine**. The students observed that under room temperature and pressure, these elements exist in noticeably different physical states: fluorine and chlorine are gases, bromine is a dense volatile liquid, and iodine is a dark, shiny solid that easily sublimes. During the laboratory demonstration, the chemist demonstrated the reactions of the elements with **water** and **sodium hydroxide solution**.The plant safety manager emphasized that while halogens and their derivatives are essential in various fields, they can pose severe health and environmental risks.The chemistry teacher tasked the students with writing a comprehensive scientific report covering the physical properties, chemical reactions, applications, hazards, and mitigations associated with halogens.

ITEM five

Different oxides are used in industries such as cement, fertilisers, and ceramics, where their properties affect heating, safety, and environmental pollution. Farmers near a certain factory reported crop damage and faster corrosion of iron sheets while workers and engineers raised concerns about irritating oxide fumes and the high temperatures required to process some oxide materials. Concerned about the increasing complaints, the district science and

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environment committee organised a joint industrial inspection involving chemists, environmental officers, and Senior Six chemistry learners from neighbouring schools.

Samples of several oxides formed by Period 3 elements were collected from different industrial processes and analysed in the laboratory. The learners observed the following information:

Oxide	Na_2O	MgO	Al_2O_3	SiO_2	SO
Melting point (°C)	1275	2827	2007	1607	30

During the inspection, the learners observed that some oxides reacted vigorously with water, some dissolved only in alkaline solutions, while others contributed to acidic emissions in the atmosphere.

The district committee later requested the learners to prepare a scientific report that could help industries improve safety, reduce pollution, and select suitable oxide materials for industrial use.

Task: *As one of the Senior Six learners of chemistry.*

a) Explain why some oxides formed fumes while others require high temperatures to process.

b) Analyse and explain the observed reactions of the oxides investigated.

c) Assess the environmental, health, or industrial implications associated with the production and use of the oxides investigated, and recommend suitable safety measures.

ITEM 6

One of the manufacturing industries, has decided to process some items for use to the community. The project planner has recommended that they should use the elements iron, silicon, tin and lead as source of raw material.

The industry wants to make salts of iron among which includes compounds with formula, $Fe(CO)_4$ and $Fe(H_2O)_6^{2+}$ used in treatment of anemia and transportation of oxygen in blood respectively.

The project planner also found out that the use of silicon, tin and lead depends on variation in their physical nature and preferred to use the dioxides of the elements in his manufacture.

Furthermore, he said his concern requires having clear understanding on;

How the dioxides interact with sodium hydroxide and hydrochloric acid, Application and impact of one of the compounds of iron on health.

Task; Write a report;

(a) Involving identification, and justification for formation of the given compounds of iron, and also to help the implementer understand variation in physical nature.

(b) to help the implementer address his concerns.

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ASSESEMENT OBJECTIVE 2 (AO2)

The topics in this assessment objective include;

1. **Organic chemistry I**
2. **Organic chemistry II**
3. **Organic chemistry III**

Item one:

A local chemical manufacturing company produces solvents and organic intermediates for the textile dye industry. The plant uses **propan-1-ol** as a starting material to synthesize **propanone hydrazone**, and **benzene** to manufacture **N-ethyl-N-phenylamine**.
During production, significant quantities of unreacted starting materials remain as chemical waste. To manage this waste, the plant engineers implement two disposal/mitigation reactions: Excess **propan-1-ol** is treated with hot concentrated sulfuric acid at around $180^{0}C$ and Excess **benzene** is reacted with liquid bromine in the presence of an iron catalyst.
The plant management has requested a comprehensive technical report analyzing the suitability of the starting materials, analysis of the reactions and the reaction pathways involved in their industry operations. An evaluation of their products identifying the potential hazards and possible solutions to minimize the effects of the organic compounds used in the operations of the company.

TASK:

As an A-Level Chemistry student, analyze the provided information and write a detailed technical report to address the aspects in brief provided.

Item two

A small beverage and flavoring processing plant operating in the Namanve Industrial Park produces aromatic concentrates and additives for local fruit juice processors. In their current operations, the plant synthesizes two main chemical products:

1. An ester with a fruity odor, produced by heating propanoic acid with ethanol in the presence of concentrated sulfuric acid.
2. A bright yellow-orange solid formed by reacting propanone with acidified 2,4-dinitrophenylhydrazine (2,4-DNPH).

Because of its vibrant color, some members of the production team have proposed blending the yellow-orange solid directly into fruit juices as a low-cost food coloring agent.
Additionally, the company recently acquired a bulk supply of ethanol from local distillers. To minimize operational costs, management plans to use this ethanol as a starting material to synthesize two of their major raw materials: **propanone** and **propanoic acid**. However, the plant engineers need to determine whether these conversion pathways are chemically viable. The manager is also concerned about the potential health, safety, and environmental risks associated with handling these chemical compounds and releasing new products to the market.

TASK:

As a Senior Six Chemistry student, analyze the provided information and prepare a comprehensive scientific write-up for the company management. Your write out should also

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give an assessment the physical properties of the the raw materials used by the plant. In addition evaluate the suitability of the products of the processing plant.

Item three:

A pharmaceutical and agrochemical manufacturing company operating in Kampala uses various halogenated and aromatic organic compounds as primary feedstock for industrial synthesis.During a routine chemical store audit, the quality control team retrieved safety data sheets and physical profiles for three unidentified organic liquid raw materials:

- Compound A: An halogenated alkane containing two carbon atoms and one chlorine atom per molecule, with a relative molecular mass of approximately 64.5.
- Compound B: An aromatic hydrocarbon containing six carbon atoms per molecule, with a relative molecular mass of 78.
- Compound C: An aromatic hydrocarbon containing seven carbon atoms per molecule, with a relative molecular mass of 92.

The plant production team uses Compound B as the precursor to synthesize chlorobenzene and nitrobenzene through reactions with respective reagents. Meanwhile, the technicians after interacting with a chemist realize that in order to save the company resources, compound A can be converted to compound C while compound B can be converted to a dye that can be used by the company to color aprons for the workers. The plant engineer needs a detailed technical review analyzing the physical properties of these feedstock chemicals, evaluating the chemical conversion processes, mechanisms involved in concerting compound B to the named compounds above, and assessing safety of using the feedstock compound.

TASK:

As a Senior Six Chemistry student, analyze the provided information and write a comprehensive scientific report to assist the company's engineering team.

Item four

A local manufacturing enterprise in the Kampala Industrial Area intends to set up a plant to produce lightweight packaging materials and synthetic textile fibers to meet growing urban demand. The research department proposes utilizing organic chemical synthesis to produce two essential industrial polymers: **Polystyrene** (an addition polymer) and a **Polyester** such as Polyethene Terephthalate (PET, a condensation polymer) from chloroethane.

To secure operational clearance from the National Environment Management Authority (NEMA) and funding from technical partners, the enterprise must submit a comprehensive technical proposal. The proposal needs to outline the steps for the synthesis of the polymers from the identified precursor, reaction mechanisms for the formation of polystyrene from its monomer and formation of chloroethane from ethane. The proposal should include the properties of the polymers of interest and an assessment of the applications, impacts of the polymers and actionable risk-mitigation strategies.

As a consulting chemist, you have been contracted to draft this technical document.

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Item five

An agrochemical manufacturing plant located at Luzira Industrial Area synthesizes a fumigant used in post-harvest grain preservation. The active ingredient in this formulation is 1,2-dichloroethane and is dissolved in high-purity propanone as the industrial solvent to completely dissolve the pesticide.

During a routine supply audit, a new quality assurance intern raised two management questions to the technical team.

- The intern noted that instead of disposing of the raw expired batches of 1,2-dichloroethane pesticide, the factory could study how it could be chemically transformed into ethyne.
- The intern discovered an unlabeled container in the solvent storage bay. To verify whether it was the required propanone solvent, they suggested sampling the liquid and testing how it reacts with Brady's reagent (2,4-dinitrophenylhydrazine).

Furthermore, due to shortages in supply, the research division is investigating green chemistry pathways to make both components locally available from raw materials. They want to map out how the pesticide (1,2-dichloroethane) can be derived from chloroethane, and the solvent (propanone) can be efficiently produced starting from ethanol.

Understanding the boiling points of the components and their raw materials, as well as evaluating the use of propanone as the industrial solvent, is important.

Task: Make a report with guidance on the issues raised.

END

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